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Dataset Open

Standard Task Graph (STG) Dataset with JSON Conversions for Workflow Scheduling in Heterogeneous High Performance Computing (HPC) Systems

Sharma, Aasish Kumar (Researcher)[1, 2]

Show affiliations

Contributors

Producer: KASAHARA, Hiroshi[3, 4]

Supervisor: Kunkel, Julian[1, 2]

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This dataset extends the Standard Task Graph (STG) benchmark suite while working under supervision of Prof. Dr. Julian Kunkel where the STG dataset originally developed by Prof. Dr. Hiroshi Kasahara and collaborators at Waseda University.

The dataset converts classical STG task graph files into machine-readable JSON workflow descriptions suitable for modern workflow scheduling research in heterogeneous high-performance computing (HPC) environments.

Contributor:

- Prof. Dr. Julian Kunkel (Supervisor)

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STG To JSON Conversion:

- Aasish Kumar Sharma, University of Göttingen

Keywords:

- Standard Task Graph
- HPC Scheduling
- Workflow Scheduling
- Task Graph Dataset
- Heterogeneous Computing

Files

README.md

Standard Task Graph (STG) Dataset with JSON Conversions for Workflow Scheduling in Heterogeneous HPC Systems

Dataset Summary

This dataset extends the Standard Task Graph (STG) benchmark suite by providing benchmark-ready JSON workflow representations and system configurations for workflow scheduling research in homogeneous and heterogeneous HPC environments. The dataset is intended for optimization, heuristic, and learning-based scheduling methods.

Supported Tasks

- Standard Task Graphs (STG)
- Workflow scheduling
- Heterogeneous HPC resource allocation
- Constraint-aware mapping and scheduling

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Dataset: https://www.kasahara.cs.waseda.ac.jp/schedule/stgarc_e.html (URL)

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Conference paper: 10.1109/COMPSAC65507.2025.00341 (DOI)

Dates

Created

2026-03

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Software

Repository URL

https://github.com/AasishKumarSharma/grapheonrl-benchmark

Programming language

Python

Development Status

Active

References

- bibtex @dataset{sharma2026stgjson, author = {Aasish Kumar Sharma}, title = {Standard Task Graph (STG) Dataset with JSON Conversions for Workflow Scheduling in Heterogeneous HPC Systems}, year = {2026}, publisher = {Zenodo}, doi = {10.5281/zenodo.18927122} }

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Keywords and subjects

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Thesis

University of Gottingen (Program of Computer Science), PhD

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